NEWTON SCHOOL

PRE-CALCULUS | HOMEWORK

Sequences & Series: Arithmetic, Geometric & Infinite

Student: Justin Koo **Date:** 2026-04-26 **Due:** 2026-05-03

Standards: *CCSS.MATH.CONTENT.HSF-BF.A.2 — Write arithmetic and geometric sequences both recursively and explicitly · HSF-IF.A.3 — Recognize that sequences are functions defined on natural numbers.*

📐 Formula Sheet

Type	nth term	Sum
Arithmetic	$t_n = t_1 + (n-1)\,d$	$S_n = \frac{n}{2}(t_1 + t_n) = \frac{n}{2}\big(2t_1 + (n-1)d\big)$
Geometric	$t_n = t_1 \cdot r^{n-1}$	$S_n = t_1 \cdot \frac{r^n - 1}{r - 1} \quad (r \neq 1)$
Infinite Geometric	—	$S_\infty = \frac{t_1}{1 - r} \quad (\,\lvert r\rvert < 1\,)$

EXAMPLE 1 *Arithmetic — finding a term*

Given the arithmetic sequence with $t_1 = 7$ and $t_2 = 21$, find t_{20}.

Step 1. Common difference: $d = t_2 - t_1 = 21 - 7 = 14$.

Step 2. Apply $t_n = t_1 + (n-1)\,d$:

$$t_{20} = 7 + (20-1)(14) = 7 + 19 \times 14 = 7 + 266 = 273$$

$$\boxed{t_{20} = 273}$$

EXAMPLE 2 *Geometric — finding a term*

Given the geometric sequence with $t_1 = 3$ and $t_2 = 6$, find t_{10}.

Step 1. Common ratio: $r = \frac{t_2}{t_1} = \frac{6}{3} = 2$.

Step 2. Apply $t_n = t_1 \cdot r^{n-1}$:

$$t_{10} = 3 \cdot 2^{10-1} = 3 \cdot 2^9 = 3 \cdot 512 = 1536$$

$$\boxed{t_{10} = 1536}$$

Part A · Arithmetic Sequences

Directions: *Find the indicated term of each arithmetic sequence. Show your work.*

1. Given $t_1 = 12$ and $t_2 = 18$, find t_{20}.

Answer: $t_{20} =$ _____

2. Given $t_3 = 10$ and $t_5 = 18$, find t_{40}.

Answer: $t_{40} =$ _____

3. Given $t_1 = 88$ and $t_3 = 80$, find t_{50}.

Answer: $t_{50} =$ _____

Part B · Geometric Sequences

Directions: *Find the indicated term of each geometric sequence. Show your work.*

4. Given $t_1 = 5$ and $t_2 = 10$, find t_{10}.

Answer: $t_{10} =$ _____

5. Given $t_2 = 27$ and $t_4 = 3$, find t_8.

Answer: $t_8 =$ _____

6. Given $t_1 = 2$ and $t_2 = 2^{3/2}$, find t_{13}.

Answer: $t_{13} =$ _____

Part C · Series and Sums

Directions: *Find the specified sum. Show your work.*

7. Find S_{10} of the arithmetic sequence: $5,\ 12,\ 19,\ 26,\ \ldots$

Answer: $S_{10} =$ _____

8. Find S_8 for the geometric sequence with $t_1 = 8$ and $r = \dfrac{1}{2}$.

Answer: $S_8 =$ _____

Part D · Infinite Geometric Series

Directions: *Find the sum of each infinite geometric series, or state that it diverges.*

9. $1 + \frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \cdots$

Answer: $S_\infty =$ _____

10. Convert the repeating decimal $0.\overline{45}$ to a fraction using an infinite geometric series.

Answer: $0.\overline{45} =$ _____

📋 **Submission Checklist**

- Show all work — partial credit for setup, full credit for clean derivations.
- **Box every final answer.**
- State which formula you're using at the start of each problem.
- For Part D, check $|r| < 1$ before applying the infinite-sum formula.

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